

SOLUTIONS

Name: _____

Date: _____



Year 10 Mathematics Mainstream

Test 1, 2014

Topics – Surds & Indices

Total Time:

60 minutes

Weighting:

6% of the year.

Equipment

Allowed:

1/2 page notes (A4 one side)

60
56
= _____ %

CALCULATOR FREE TEST

1. [2 marks: 1, 1]

Write these surds in simplest form:

a) $\sqrt{28}$

$$= \sqrt{4 \sqrt{7}}$$

$$= 2\sqrt{7}$$

b) $\sqrt{300}$

$$= \sqrt{100 \sqrt{3}}$$

$$= 10\sqrt{3}$$

2. [2 marks: 1, 1]

Simplify each of the following

a) $2\sqrt{7} - 6\sqrt{7} + 9\sqrt{7}$

$$= 5\sqrt{7}$$

b) $2\sqrt{5} + 8\sqrt{14} - 8\sqrt{5} - 6\sqrt{14}$

$$= -6\sqrt{5} + 2\sqrt{14}$$

3. [3 marks: 1, 2]

Simplify each of the following

a) $\sqrt{3} + \sqrt{20}$

$$= \sqrt{3} + \sqrt{4 \sqrt{5}}$$

$$= \sqrt{3} + 2\sqrt{5}$$

b) $\sqrt{8} + \sqrt{24} + \sqrt{18} + \sqrt{54}$

$$= \sqrt{4 \sqrt{2}} + \sqrt{4 \sqrt{6}} + \sqrt{9 \sqrt{2}} + \sqrt{9 \sqrt{6}}$$

$$= 2\sqrt{2} + 2\sqrt{6} + 3\sqrt{2} + 3\sqrt{6}$$

$$= 5\sqrt{2} + 5\sqrt{6}$$

working ①
answer ①

4. [3 marks: 1, 1, 1]

Simplify each of the following

a) $\sqrt{3} \times \sqrt{3}$

$$= 3$$

b) $\sqrt{5} \times \sqrt{12}$

$$= \sqrt{60} \quad \left(\frac{1}{2}\right)$$

$$= \sqrt{4 \sqrt{15}}$$

$$= 2\sqrt{15} \quad \left(\frac{1}{2}\right)$$

c) $5\sqrt{7} \times 2\sqrt{8}$

$$= 10\sqrt{56} \quad \left(\frac{1}{2}\right)$$

$$= 10 \times \sqrt{4 \sqrt{14}}$$

$$= 10 \times 2\sqrt{14}$$

$$= 20\sqrt{14} \quad \left(\frac{1}{2}\right)$$

5. [2 marks: 1, 1]
Simplify each of the following

a) $\sqrt{27} \div \sqrt{2}$

$$= \frac{\sqrt{9}\sqrt{3}}{\sqrt{2}}$$

$$= \frac{3\sqrt{3}}{\sqrt{2}}$$

b) $\frac{18\sqrt{75}}{3\sqrt{3}}$

$$= \frac{6\sqrt{25}\sqrt{3}}{\sqrt{3}}$$

$$= 6 \times 5$$

$$= 30$$

6. [4 marks: 1, 3]
Simplify each of the following:

a) $3(5 + \sqrt{3})$

$$= 15 + 3\sqrt{3}$$

b) $(4 + \sqrt{5})(\sqrt{2} - \sqrt{8})$

$$= 4\sqrt{2} - 4\sqrt{8} + \sqrt{10} - \sqrt{40}$$

working (2)
answer (1)

$$= 4\sqrt{2} - 4\sqrt{4}\sqrt{2} + \sqrt{10} - \sqrt{4}\sqrt{10}$$

$$= 4\sqrt{2} - 8\sqrt{2} + \sqrt{10} - 2\sqrt{10}$$

$$= -4\sqrt{2} - \sqrt{10}$$

7. [4 marks: 1, 1, 1, 1]
Evaluate

a) $\frac{3^6}{3^3} = 3^{6-3} = 3^3$
 $= 27$

b) 2^{-4}
 $= \frac{1}{2^4} = \frac{1}{16}$

c) $4^0 + 2(x + b^2)^0$

$$= 1 + 2 \times 1$$

$$= 1 + 2$$

$$= 3$$

d) $(2^3)^2$

$$= 2^6$$

$$= 64$$

8. [4 marks: 1, 1, 1, 1]
Simplify each of the following.

a) $5x^4 \times 2x \times 3x^3$

$$= 30x^{4+1+3}$$

$$= 30x^8$$

b) $\frac{20x^6}{5x^3}$

$$= 4x^{6-3}$$

$$= 4x^3$$

c) $4x^0$

$$= 4 \times 1$$

$$= 4$$

d) $\left(\frac{a^3}{b^5}\right)^4$

$$= \frac{a^{3 \times 4}}{b^{5 \times 4}}$$

$$= \frac{a^{12}}{b^{20}}$$

- 9 8. [6 marks: 1, 1, 2, 2]
Simplify each of the following.

a) $3x^3 \times 4x^2y^4 \times 2xy^5$

$$= 24x^{3+2+1}y^{4+5}$$

$$= 24x^6y^9$$

b) $(2x^2y^3)^2$

$$= 2^2x^{2 \times 2}y^{3 \times 2}$$

$$= 4x^4y^6$$

c) $\frac{80x^8y^{16}}{12x^2y^9}$

$$= \frac{20x^{8-2}y^{16-9}}{3}$$

working ①
answer ①

$$= \frac{20x^6y^7}{3}$$

d) $(x^5y^2)^4 \times (2x^2y^6)^3$

$$= x^{5 \times 4}y^{2 \times 4} \times 2^3x^{2 \times 3}y^{6 \times 3}$$

$$= x^{20}y^8 \times 8x^6y^{18}$$

working ①
answer ①

$$= 8x^{20+6}y^{8+18}$$

$$= 8x^{26}y^{26}$$

- 10 8. [4 marks: 2, 2]
Simplify each of the following.

e)
$$\frac{5x^4 \times 2x \times \frac{1}{2}y^9}{15x^2y^3}$$

$$= \frac{5x^5y^9}{15x^2y^3}$$

working ①
answer ①

$$= \frac{1x^{5-2}y^{9-3}}{3}$$

$$= \frac{x^3y^6}{3}$$

f)
$$\frac{40x^9}{y} \div \frac{12x}{2y^4}$$

$$= \frac{40x^9}{y} \times \frac{2y^4}{6 \cancel{12}x}$$

$$= \frac{40x^9y^4}{\cancel{6}xy}$$

working ①
answer ①

$$= \frac{20x^{9-1}y^{4-1}}{3}$$

$$= \frac{20x^8y^3}{3}$$

- 11 9. [6 marks: 1, 1, 2, 2]
Simplify each of the following. Write your answers with positive indices.

a) x^{-6}

$$= \frac{1}{x^6}$$

b) $\frac{7a^{-2}}{4b^{-4}}$

$$= \frac{7b^4}{4a^2}$$

- 12 10. [4 marks: 2, 2]

Simplify each of the following. Write your answers with positive indices.

c) $7x^7y^{-2} \times 8x^{-5}y^{-4}$

$$= 56x^{7+(-5)}y^{-2+(-4)}$$

$$= 56x^2y^{-6}$$

$$= \frac{56x^2}{y^6}$$

working ①
answer ①

d) $\frac{m^{-2}n^{-5}}{m^{-8}n^3}$

$$= m^{-2-(-8)}n^{-5-3}$$

$$= m^6n^{-8}$$

$$= \frac{m^6}{n^8}$$

working ①
answer ①

10. [4 marks: 2, 2]

12 Simplify each of the following. Write your answers with positive indices

a)

$$\begin{aligned} & \left(\frac{3m^{-6}}{p^4} \right)^{-3} \\ &= \frac{3^{-3} m^{-6 \times -3}}{p^{4 \times -3}} \\ &= \frac{3^{-3} m^{18}}{p^{-12}} \\ &= \frac{m^{18} p^{12}}{3^3} \end{aligned}$$

working ①
answer ①

b)

$$\begin{aligned} & \frac{3a^4}{(2x^{-2})^{-3}} \\ &= \frac{3a^4}{2^{-3} x^{-2 \times -3}} \\ &= \frac{3a^4}{2^{-3} x^6} \\ &= \frac{3a^4 \times 2^3}{x^6} \\ &= \frac{24a^4}{x^6} \end{aligned}$$

working ①
answer ①

13

11. [6 marks: 3, 3]

Simplify each of the following. Write your answers with positive indices.

a)

$$\begin{aligned} & \left(\frac{2b^2}{5b^{-4}} \right)^{-2} \\ &= \frac{2^{-2} b^{2 \times -2}}{5^{-2} b^{-4 \times -2}} \\ &= \frac{2^{-2} b^{-4}}{5^{-2} b^8} \\ &= \frac{5^2 b^{-4-8}}{2^2} \\ &= \frac{25 b^{-12}}{4} \\ &= \frac{25}{4b^{12}} \end{aligned}$$

working ②
answer ①

b)

$$\begin{aligned} & \left(\frac{-3m^0 n^5}{n^{-2}} \right)^3 \\ &= \frac{(-3)^3 m^{0 \times 3} n^{5 \times 3}}{n^{-2 \times 3}} \\ &= \frac{-27 m^0 n^{15}}{n^{-6}} \\ &= -27 n^{15-(-6)} \\ &= -27 n^{21} \end{aligned}$$

working ①
answer ①

13. [3 marks]

14 Simplify each of the following. Write your answers with positive indices.

a) $\left(\frac{4a^6}{b^5}\right)^3 \div \left(\frac{4^2a^2}{4b^{-2}}\right)^4$

$$= \frac{4^3 a^{6 \times 3}}{b^{5 \times 3}} \times \frac{4^4 b^{-2 \times 4}}{4^{2 \times 4} a^{2 \times 4}}$$

$$= \frac{4^3 a^{18}}{b^{15}} \times \frac{4^4 b^{-8}}{4^8 a^8}$$

$$= \frac{4^{3+4} a^{18-8} b^{-8-15}}{4^8}$$

$$= \frac{4^7 a^{10} b^{-23}}{4^8}$$

$$= \frac{4^{-1} a^{10}}{b^{23}}$$

$$= \frac{a^{10}}{4b^{23}}$$

working ②

answer ①

14. [3 marks]

15 Evaluate the following for $x = 4$

$$3x^{-2} \times \left(\frac{2}{x}\right)^4 \div \frac{3x}{(x^{-2})^{-3}}$$

$$= 3(4)^{-2} \times \left(\frac{2}{4}\right)^4 \div \frac{3 \times 4}{(4^{-2})^{-3}}$$

$$= \frac{3 \times 4^{-2}}{1} \times \frac{2^4}{4^4} \times \frac{4^6}{3 \times 4}$$

$$= \frac{4^{-2+6}}{4^{4+1}} \times 2^4$$

$$= \frac{4^4 \times 2^4}{4^5} = 4^{-1} \times 2^4 = \frac{2^4}{4} = \frac{16}{4} = 4$$

working ②

answer ①

~END OF TEST~